

Blake Grudzien

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Experience

Level Home

May 2025 – August 2025

Software Engineer Intern

Redwood City, CA

- Architected serverless telemetry pipeline processing millions of IoT events using AWS (Lambda, SQS, S3, Athena), reducing data loss from 30% to <1% while replacing a legacy system with reliability and cost issues
- Built automated Slack notification system alerting the firmware team of parsing failures in real-time, streamlining onboarding of new telemetry events to production devices
- Designed Dead Letter Queue system enabling failed telemetry files to be reprocessed after parser updates, eliminating permanent data loss from firmware changes and reducing coordination overhead between the firmware and data teams
- Migrated telemetry storage from row-based JSON to Parquet, letting Athena scan only queried columns
- Developed plug-and-play parser registry in Go supporting multiple device types (locks, bridges, doorbells), enabling new products to be onboarded without pipeline refactoring

Brooksee

June 2024 – October 2024

Software Engineer Intern

Salt Lake City, UT (Hybrid)

- Built custom Garmin Connect IQ data field (MonkeyC) paired with a Kotlin Android app, designed to stream live pace, distance, placement, and gun-time gaps to runners' watches during official Brooksee races (5k-10k runners)
- Engineered Firestore-to-Bluetooth telemetry pipeline with server-synced timing and fault-tolerant data parsing
- Derived each runner's elapsed time from the race's absolute Unix start time rather than a local timer, so runners joining at irregular intervals after the gun still showed correct official time despite device clock drift

Education

Emory University

Atlanta, GA

B.S. Computer Science, Minor in Mathematics | GPA: 3.66/4.00

May 2026

- Coursework: Data Structures & Algorithms, Machine Learning, AI, Database Systems, Mathematical Statistics, Linear Algebra, Data Mining, Machine-Level Programming, Discrete Mathematics

Technical Skills

Languages: Python, Go, Java, JavaScript, Kotlin, SQL, C++/C, HTML, CSS, R

Frameworks & Libraries: Flask, React, SvelteKit, Streamlit, Spring, Scikit-Learn, pandas, NumPy, SQLAlchemy

DevOps & Cloud: AWS (Lambda, SQS, S3, Athena), Git, Docker, PostgreSQL/pgvector, MongoDB, Redis, Firestore

Other: RAG, vector embeddings, LLM tool-calling, REST APIs, Agile/Scrum, Linux, Prompt Engineering

Projects

Chess RAG Research Assistant ([Live](#) | [Github](#)) | *Python, Claude API, Postgres/pgvector, Stockfish* 2026 – Present

- Built a hybrid retrieval system over 127k games and 10M moves (1610–2023), routing questions across deterministic SQL, pgvector semantic search over 469k embedded annotation chunks, and Stockfish engine analysis via native LLM tool-calling
- Required engine grounding before any model claim about move quality, constraining the LLM to attribute answers to retrieved stats, prose, or evaluations rather than its own judgment

CHOA Chatbot ([Github](#)) | *Python, React, MongoDB*

Jan 2024 – May 2024

- Built AI-powered chatbot for the Children's Hospital of Atlanta in a 7-person team, training resident doctors on pediatric airway management protocols for respiratory emergencies
- Delivered weekly stakeholder updates and adapted iteratively to changing clinical requirements using Agile methods

Bird Strike Analysis ([Live](#) | [Github](#)) | *SvelteKit, D3.js, TypeScript*

Feb 2026 – May 2026

- Built an interactive data-driven article analyzing 300k+ FAA bird strike records, featuring animated geographic maps, linked charts, and a species heatmap, deployed via GitHub Actions
- Structured the analysis as a narrative for non-technical readers, translating raw incident data into a clear argument about airport and airline risk factors

Leadership

Vice President

Oct 2023 – Sep 2024

Alpha Tau Omega Fraternity, Emory University (100+ members)

Atlanta, GA

- Led chapter to 3rd highest GPA among all 136 national ATO chapters; served as head of disciplinary process, mediating conflicts across 100+ members
- Delivered presentations on safety protocols and organizational values to general membership

Additional

Chess: 2209 chess.com elo (99.9 percentile) — **Languages:** Intermediate Spanish — **Interests:** Tennis, Rock Climbing, Soccer